Quantitative Wasserstein-1 Contraction for Overdamped Langevin Dynamics under a Bounded Lipschitz Perturbation of a Strongly Convex Potential: An Explicit Rate via Reflection Coupling

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Abstract

We prove an explicit, closed-form Wasserstein contraction rate for overdamped Langevin dynamics $dX_t = -\nabla U(X_t)\,dt + \sqrt{2}\,dB_t$ in $\mathbb{R}^d$, in the specific but widely used setting where the potential decomposes as $U = U_0 + V$ with U_0 being m-strongly convex and V a $C^{1,1}$ perturbation with $\|\nabla V\|_\infty \le K$. In the subcritical regime $K < \sqrt{m}$, we construct an explicit concave distance function f, a one-parameter power law with exponent determined by the dimensionless ratio $K/\sqrt{m}$, and show that two solutions coupled by the Eberle reflection coupling contract at rate

$$c^\star \;=\; m\big(1 - K/\sqrt{m}\big) \;=\; m - K\sqrt{m}$$

in the corresponding Wasserstein-1 metric. The rate is dimension-free, independent of $\|V\|_\infty$, and depends on the perturbation only through its gradient sup-norm K. The construction specialises Eberle's variational contraction theorem [5] to a numerically concrete bound that can be inserted directly into discretisation error analyses for the unadjusted Langevin algorithm (ULA) and into propagation-of-chaos estimates for mean-field systems.

1 Introduction

1.1 Setting

Consider the overdamped Langevin diffusion on $\mathbb{R}^d$,

$$dX_t \;=\; -\nabla U(X_t)\,dt \;+\; \sqrt{2}\,dB_t, \tag{1}$$

where $(B_t)_{t\ge 0}$ is a standard Brownian motion in $\mathbb{R}^d$ and $U : \mathbb{R}^d \to \mathbb{R}$ is a C^2 potential. We focus on the case where U admits the decomposition

$$U \;=\; U_0 + V, \tag{2}$$

with U_0 strongly convex and V a bounded-gradient perturbation. This structure arises pervasively in Bayesian posteriors with bounded-feature log-likelihoods, Gibbs measures in statistical mechanics with a bounded external field, and penalised sampling objectives in machine learning. The perturbation V is not assumed to be convex, and in general the full potential U is not convex either: the SDE (1) may exhibit metastable wells induced by V.

For such dynamics, exponential contraction in Wasserstein distance is the central quantitative tool. It controls: the relaxation rate of the law of X_t towards the invariant measure $\pi \propto e^{-U}$

(Bakry–Émery [14], Otto–Villani [16], Bakry–Gentil–Ledoux [15]), the discretisation error of the unadjusted Langevin algorithm (ULA) of Durmus and Moulines [2, 4, 28], the uniform-in-time propagation of chaos rate for the associated mean-field particle system (Sznitman [11], Eberle–Guillin–Zimmer [6], Lacker [34, 35], Guillin–Liu–Wu–Zhang [38]), and the Lipschitz stability of the semigroup under further perturbations.

1.2 Prior art

The case in which U itself is strongly convex is classical: synchronous coupling yields W_2-contraction at the convexity rate (von Renesse–Sturm [12], with the underlying functional-inequality picture due to Bakry–Émery [14] and developed in [15, 17, 18]). For non-convex U satisfying a *distant-dissipativity* condition, the Lindvall–Rogers [8] reflection coupling combined with Chen–Wang's spectral-gap technique [19] was first refined into an exponential contraction by Eberle [20, 5], who constructs a carefully chosen concave distance function f and proves exponential W_1-type contraction at a rate $c > 0$ given by a variational principle involving the solution of a Sturm–Liouville problem. The rate in [5] is typically not available in closed form. Subsequent work has extended the reflection-coupling technology in several directions: Eberle, Guillin, and Zimmer [6] obtained quantitative Harris-type theorems for both diffusions and McKean–Vlasov dynamics; Eberle and Zimmer [21] extended to "sticky" couplings of diffusions with different drifts; Luo and Wang [9] gave an L^p-Wasserstein version; Wang [13] analysed the case of negatively curved drift via parallel-transport coupling; Hairer, Mattingly, and Scheutzow [43] formulated a general Harris-type framework. The Markov-chain-stability tradition descending from Meyn and Tweedie [44] and Khasminskii [45] provides the foundational well-posedness and Lyapunov technology. For the closely related Hamiltonian Monte Carlo scheme, Bou-Rabee, Eberle, and Zimmer [1] derived explicit coupling bounds under distant dissipativity, with earlier MALA mixing analysis by Bou-Rabee and Hairer [40]; for proximal Langevin under strong convexity, Durmus, Moulines, and Pereyra [3] gave explicit constants.

A parallel and complementary literature analyses the unadjusted Langevin algorithm (ULA) and its kinetic variant under entropy or KL-divergence: Cheng and Bartlett [22]; Cheng, Chatterji, Bartlett, and Jordan [23]; Vempala and Wibisono [24]; Chewi, Erdogdu, Li, Shen, and Zhang [25]; Lehec [27]; Dalalyan and Riou-Durand [29]; the recent textbook Chewi [26]; further refinements of step-size dependence include Neufeld and Xiang [47] and Zou, Gu, and Zhang [48]. The numerical-analysis foundations of the Euler–Maruyama scheme go back to Kloeden and Platen [30] and Talay [31], with ergodicity-of-discretisation results by Mattingly, Stuart, and Higham [41].

Quantitative propagation of chaos for mean-field interacting systems has seen rapid recent progress: Jabin and Wang [32] for $W^{-1,\infty}$ kernels, Bresch–Jabin–Wang [33] for singular kernels including Patlak–Keller–Segel, Lacker [34, 35] on hierarchies and entropy-method propagation, Monmarché [36] for kinetic mean-field particles, Delarue and Lacker [37], and Durmus–Eberle–Guillin–Schuh [39] on sticky nonlinear SDEs. Hairer [42] and Eldan [46] provide further structural perspectives.

Despite this well-developed machinery, there is, to the author's knowledge, no reference that carries out the Eberle construction for the specific structured class (2) and extracts a closed-form rate expressed purely in m and K. This is the gap the present note fills, in the technically simple but useful subcritical regime $K < \sqrt{m}$.

1.3 Main contributions

This paper has five results, tied together by the same explicit contraction rate $c^\star = m - K\sqrt{m}$ in the subcritical regime $K < \sqrt{m}$.

(1) Closed-form continuous-time contraction (Theorem 8). For the overdamped Langevin diffusion (1) with $U = U_0 + V$ satisfying Assumptions 1–3, any two solutions coupled via the Eberle reflection coupling satisfy

$$W_{1,f}\big(\mathrm{Law}(X_t), \mathrm{Law}(X'_t)\big) \;\le\; e^{-c^\star t}\, W_{1,f}(\mu_0, \nu_0), \qquad c^\star = m - K\sqrt{m},$$

where f is the explicit one-parameter power-law concave distance (9) with exponent $1-\beta$, $\beta := K/\sqrt{m}$.

(2) Discrete-time contraction for the unadjusted Langevin algorithm (Theorem 12). Under the additional assumption that ∇U is globally L-Lipschitz, the ULA iterates with step size $h \le h_0 := \min(m/(2L^2),\, 1/L,\, 1/c^\star)$ satisfy

$$W_{1,f}\big(\mathrm{Law}(\bar X_N), \pi\big) \;\le\; e^{-c^\star N h/2}\, W_{1,f}(\mathrm{Law}(\bar X_0), \pi) \;+\; \frac{C_1\, L^2\, \sqrt{M^\star + d}\, \sqrt{h}}{c^\star},$$

where $\pi \propto e^{-U}$ is the invariant measure, $M^\star = \mathcal{O}(d + K^2/m^2)$ is the uniform ULA second-moment bound (Lemma 15), and C_1 is an absolute constant arising from the Euler-Maruyama one-step strong error. The rate $c^\star$ is the *same* as in the continuous-time bound; the discretisation cost enters only through the additive $O(\sqrt{h})$ term.

(3) Uniform-in-time, dimension-free propagation of chaos for the mean-field particle system (Theorem 18). Let

$$dX_t^{i,N} \;=\; -\nabla U_0(X_t^{i,N})\, dt \;-\; \frac{1}{N}\sum_{j=1}^{N} \nabla_1 W(X_t^{i,N}, X_t^{j,N})\, dt \;+\; \sqrt{2}\, dB_t^i, \qquad i = 1, \dots, N,$$

be the McKean-Vlasov interacting particle system with U_0 m-strongly convex and $W : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ a pairwise interaction with $\|\nabla_1 W(\cdot, y)\|_\infty \le K$ uniformly in y. In the subcritical regime $K < \sqrt{m}$, under the additional Lipschitz-in-the-second-argument condition $L_W \le c^\star/2$,

$$\sup_{t \ge 0} W_{1,f}\big(\mathrm{Law}(X_t^{i,N}), \mu_t\big) \;\le\; \frac{C_2\, K}{\sqrt{N-1}\, c^\star},$$

where μ_t is the solution of the limiting nonlinear Fokker-Planck equation and C_2 is an explicit absolute constant. The bound is *uniform in time* and *dimension-free*; both features distinguish it from the classical Sznitman-type bounds whose constants grow as e^{Ct}.

(4) Sharpness of the exponent (Proposition 21). Within the two-parameter family of power-law concave distances

$$f_{R,\beta}(r) \;:=\; \frac{R}{1-\beta}\big[(1 + r/R)^{1-\beta} - 1\big], \qquad R > 0,\; \beta \in [0, 1),$$

the choice $(R, \beta) = (2/\sqrt{m},\, K/\sqrt{m})$ with rate $c^\star = m(1-\beta) = m - K\sqrt{m}$ is the unique maximiser of the contraction rate. No other element of this family gives a rate larger than $m - K\sqrt{m}$.

(5) Closed-form behaviour at the critical threshold (Proposition 23). As $K \uparrow \sqrt{m}$, write $\varepsilon := \sqrt{m} - K \to 0^+$. The contraction rate vanishes *linearly* in ε with explicit constant: $c^\star = \sqrt{m}\,\varepsilon$. The concave distance f converges, uniformly on compact sets, to the logarithmic distance $f_\infty(r) = R_0 \log(1 + r/R_0)$. The bi-Lipschitz constants relating $W_{1,f}$ to the Euclidean W_1 on any bounded set remain finite. Consequently the half-life of $W_{1,f}$ from any bounded coupling is $\Theta(\varepsilon^{-1})$, giving the precise quantitative form of the breakdown at the critical threshold.

Taken together, these results give a complete, quantitatively explicit picture of exponential contraction, discretisation, mean-field propagation, sharpness of the rate within the power-law family, and the critical behaviour as $K \uparrow \sqrt{m}$, for the subcritical class $U = U_0 + V$ with $K < \sqrt{m}$. The supercritical regime $K \geq \sqrt{m}$ lies outside the reach of this power-law family and is left as an open direction, discussed in Section 11.

1.4 Main result (informal)

Let U_0 be m-strongly convex and let $V \in C^{1,1}(\mathbb{R}^d)$ satisfy $\|\nabla V\|_\infty \leq K$, with $K < \sqrt{m}$. Define

$$f(r) \;:=\; \frac{R_0}{1-\beta}\Big[(1 + r/R_0)^{1-\beta} - 1\Big], \qquad R_0 := \frac{2}{\sqrt{m}}, \quad \beta := \frac{K}{\sqrt{m}} \in (0,1), \tag{3}$$

and let $\rho(x,y) := f(|x-y|)$. Then for any two solutions $(X_t)_{t\geq 0}$ and $(X'_t)_{t\geq 0}$ of (1) coupled via the Eberle reflection coupling, with initial laws μ_0, ν_0,

$$W_{1,f}\big(\mathrm{Law}(X_t), \mathrm{Law}(X'_t)\big) \;\leq\; e^{-c^\star t}\, W_{1,f}(\mu_0, \nu_0), \qquad c^\star := m - K\sqrt{m}, \tag{4}$$

where $W_{1,f}$ is the Kantorovich-Rubinstein transport distance with cost $f(|x-y|)$. Since f is bi-Lipschitz-equivalent to the Euclidean distance on any bounded set, (4) transfers (up to an explicit prefactor) to the standard W_1-distance.

The rate $c^\star = m - K\sqrt{m}$ has three features worth emphasising:

1. **Dimension-free**: it depends only on the one-dimensional quantities m and K, not on the ambient dimension.

2. **Independent of $\|V\|_\infty$**: only the gradient sup-norm $K = \|\nabla V\|_\infty$ enters the rate; the L^∞-norm of V does not. This contrasts sharply with the Holley-Stroock log-Sobolev perturbation bound, whose constant degrades as $e^{\mathrm{osc}(V)}$ [7]. The two quantities measure different things — contraction versus functional inequalities — but the qualitative dependence on V is strikingly different.

3. **Subcritical saturation at $K = 0$**: as $K \to 0$, $c^\star \to m$, recovering the sharp strongly-convex contraction rate. At the other end, as $K \nearrow \sqrt{m}$, $c^\star \searrow 0$; beyond the critical threshold, our specific concave distance fails and a different construction is needed (see Remark 28).

The proof is short and self-contained: the concave distance f is chosen so that a single algebraic identity ($h(y) \leq \sqrt{m}$ for a specific function h on $[1,\infty)$) implies the required supermartingale inequality. The identity is verified by a residual analysis: $h(1) = \sqrt{m}$, $h'(1) = 0$, and h is strictly decreasing on $[1,\infty)$ via the sign of its bracketed linear factor.

1.5 Outline

Section 2 fixes notation and states the assumptions. Section 3 recalls the reflection coupling and derives the one-dimensional SDE for $r_t := |X_t - X'_t|$. Section 4 contains the construction of the concave distance f, the verification of the supermartingale inequality (Lemma 7), and the proof of the main theorem (Theorem 8). Section 5 records corollaries for the Euclidean Wasserstein metric and the second-moment of the coupled process. Section 6 proves the discrete-time ULA bound (Theorem 12). Section 7 establishes uniform-in-time propagation of chaos (Theorem 18). Section 8 proves sharpness of the exponent (Proposition 21). Section 9 characterises the behaviour at the critical threshold $K \uparrow \sqrt{m}$ (Proposition 23). Section 10 reports numerical experiments validating the theoretical contraction rate on a Bayesian logistic regression posterior. Section 11 collects remarks. Reproducible numerical and symbolic verification code is provided in the supplementary directory `code/`.

2 Setup and assumptions

Assumption 1. $U_0 \in C^2(\mathbb{R}^d)$ and there exists $m > 0$ such that $\nabla^2 U_0(x) \succeq mI$ for all $x \in \mathbb{R}^d$. Equivalently,

$$\langle \nabla U_0(x) - \nabla U_0(y),\, x - y \rangle \;\geq\; m\,|x-y|^2\,, \qquad \forall x, y \in \mathbb{R}^d.$$

Assumption 2. $V \in C^{1,1}(\mathbb{R}^d)$. That is, V is continuously differentiable and ∇V is Lipschitz. Moreover, there exists $K \geq 0$ such that

$$\|\nabla V(x)\| \leq K \qquad \forall x \in \mathbb{R}^d.$$

Assumption 3. $K < \sqrt{m}$.

Under Assumptions 1–2, $U := U_0 + V$ is $C^{1,1}$ and ∇U is monotone plus a bounded-gradient perturbation; the SDE (1) admits a unique strong, non-explosive solution for every initial condition by standard well-posedness theory for SDEs with locally Lipschitz drift satisfying a one-sided monotone growth condition (see, e.g., [10, Thm. 2.4.1] or Khas'minskii's criterion).

We write $\mu_t := \mathrm{Law}(X_t)$ for the marginal law at time t.

The following elementary observation, which we call *distant dissipativity* (following Eberle), is the only consequence of Assumptions 1–2 we shall use.

Lemma 4. *Under Assumptions 1–2, for all $x, y \in \mathbb{R}^d$,*

$$\langle \nabla U(x) - \nabla U(y),\, x - y \rangle \;\geq\; m\,|x-y|^2 \;-\; 2K\,|x-y|\,.$$

In particular, for $|x-y| \geq R_1 := 2K/m$, the left-hand side is non-negative and the dynamics contract synchronously.

Proof. By Assumption 1, $\langle \nabla U_0(x) - \nabla U_0(y), x-y\rangle \geq m\,|x-y|^2$. For the perturbation, by the triangle inequality and Cauchy-Schwarz applied term by term,

$$|[|\,\big]\langle \nabla V(x) - \nabla V(y),\, x - y \rangle \;\leq\; \big(\|\nabla V(x)\| + \|\nabla V(y)\|\big)\,|x-y| \;\leq\; 2K\,|x-y|\,.$$

Summing the two bounds gives the claim. The threshold $R_1 = 2K/m$ is where $mr^2 = 2Kr$. □

3 Reflection coupling

We recall the reflection coupling of [8, 5]. Given $(X_t)_{t\geq 0}$ a strong solution of (1) driven by a Brownian motion (B_t), define a companion process (X'_t) on the same probability space by

$$dX'_t \;=\; -\nabla U(X'_t)\,dt \;+\; \sqrt{2}\big(I - 2e_t e_t^\top\big)dB_t, \qquad e_t := \frac{X_t - X'_t}{|X_t - X'_t|} \tag{5}$$

for $t < \tau$, where $\tau := \inf\{t \geq 0 : X_t = X'_t\}$ is the coupling time, and $X'_t \equiv X_t$ for $t \geq \tau$. The matrix $I - 2e_t e_t^\top$ is an orthogonal reflection: X'_t is driven by the same magnitude of noise as X_t, but with the component along e_t flipped. Standard results [8] ensure that (5) admits a unique strong solution up to τ and that (X'_t) is marginally a solution of (1).

Lemma 5. *Let $r_t := |X_t - X'_t|$ for $t < \tau$. Then there exists a one-dimensional standard Brownian motion $(W_t)_{t\geq 0}$ (on the same probability space) such that*

$$dr_t \;=\; -e_t \cdot \big(\nabla U(X_t) - \nabla U(X'_t)\big)\,dt \;+\; 2\sqrt{2}\,dW_t, \qquad t < \tau. \tag{6}$$

Moreover, the drift satisfies

$$-e_t \cdot \big(\nabla U(X_t) - \nabla U(X'_t)\big) \;\leq\; 2K - m\,r_t. \tag{7}$$

Proof. From (1) and (5),

$$d(X_t - X'_t) \;=\; \big[\nabla U(X'_t) - \nabla U(X_t)\big]\,dt \;+\; 2\sqrt{2}\,e_t(e_t^\top dB_t).$$

Let $dW_t := e_t^\top dB_t$, which is a standard one-dimensional Brownian motion by Lévy's characterisation (e_t is a unit vector adapted to the filtration generated by B, so $\int_0^t e_s^\top dB_s$ is a martingale with quadratic variation t). Then

$$d(X_t - X'_t) \;=\; \big[\nabla U(X'_t) - \nabla U(X_t)\big]\,dt \;+\; 2\sqrt{2}\,e_t\,dW_t.$$

By Itô's formula applied to $|X_t - X'_t|^2$, valid on the open set $\{r > 0\}$ (equivalently, for $t < \tau$), a standard calculation gives

$$dr_t^2 = -2\langle X_t - X'_t,\, \nabla U(X_t) - \nabla U(X'_t)\rangle\,dt \;+\; 8\,dt \;+\; 4\sqrt{2}\,r_t\,dW_t.$$

Applying Itô's formula next to $r = \sqrt{r^2}$ on $\{r > 0\}$ (where $u \mapsto \sqrt{u}$ is smooth), and using $d\langle r^2\rangle_t = 32\,r_t^2\,dt$,

$$dr_t = \frac{1}{2r_t}\,dr_t^2 - \frac{1}{8r_t^3}\,d\langle r^2\rangle_t = -e_t \cdot (\nabla U(X_t) - \nabla U(X'_t))\,dt + \frac{4}{r_t}\,dt - \frac{4}{r_t}\,dt + 2\sqrt{2}\,dW_t,$$

in which the two $4/r_t$ terms cancel exactly, yielding (6). The drift bound (7) is an immediate consequence of Lemma 4, since $e_t \cdot (X_t - X'_t) = r_t$ and so

$$-e_t \cdot (\nabla U(X_t) - \nabla U(X'_t)) = -\frac{\langle X_t - X'_t, \nabla U(X_t) - \nabla U(X'_t)\rangle}{r_t} \leq \frac{2Kr_t - mr_t^2}{r_t} = 2K - mr_t.$$

□

4 The concave distance and the main theorem

4.1 The concave distance

Under Assumption 3, set

$$\beta \;:=\; \frac{K}{\sqrt{m}} \;\in\; [0,1), \qquad R_0 \;:=\; \frac{2}{\sqrt{m}}, \qquad c^\star \;:=\; m(1-\beta) \;=\; m - K\sqrt{m}, \tag{8}$$

and define

$$f(r) \;:=\; \frac{R_0}{1-\beta}\Big[\big(1 + r/R_0\big)^{1-\beta} - 1\Big], \qquad r \ge 0. \tag{9}$$

Lemma 6. *The function f defined in (9) is C^∞ on $[0,\infty)$, satisfies $f(0)=0$, $f'(0)=1$, f is strictly concave and strictly increasing, $f(r)\to\infty$ as $r\to\infty$, and*

$$f'(r) = \big(1 + r/R_0\big)^{-\beta}, \qquad f''(r) = -\frac{\beta}{R_0}\big(1 + r/R_0\big)^{-\beta-1}.$$

In particular, f is bi-Lipschitz equivalent to the identity on any bounded interval: for $r \in [0,R]$,

$$(1 + R/R_0)^{-\beta} \cdot r \;\le\; f(r) \;\le\; r,$$

so $f(r)/r \in [(1+R/R_0)^{-\beta},\, 1]$.

Proof. All claims follow from direct differentiation. Concavity is immediate since $f'' < 0$. The lower bound $f(r) \ge (1+R/R_0)^{-\beta} r$ on $[0,R]$ follows from the mean value theorem: for $r \in [0,R]$, $f(r) = \int_0^r f'(s)\,ds \ge r \cdot \inf_{s\in[0,r]} f'(s) = r \cdot (1+r/R_0)^{-\beta} \ge r \cdot (1+R/R_0)^{-\beta}$. The upper bound $f(r) \le r$ follows from $f'(s) \le f'(0) = 1$. □

4.2 The key algebraic identity

Lemma 7 (Key inequality)**.** *Under Assumptions 1–3, with f, β, $c^\star$, and R_0 as in (8)–(9), the function f satisfies, for every $r \ge 0$,*

$$(2K - mr)\, f'(r) \;+\; 4\, f''(r) \;+\; c^\star\, f(r) \;\le\; 0, \tag{10}$$

with equality if and only if $r = 0$.

Proof. Substitute f', f'' from Lemma 6 and set $y := 1 + r/R_0 \in [1,\infty)$, so that $r = R_0(y-1)$ and $2K - mr = 2K + mR_0 - mR_0\, y$. The left-hand side of (10) becomes, after algebra (and using $c^\star = m(1-\beta)$ to cancel the $y^{1-\beta}$ terms),

$$\text{LHS} \;=\; \big(2K + mR_0\big)\, y^{-\beta} \;-\; \frac{4\beta}{R_0}\, y^{-\beta-1} \;-\; mR_0 \;=\; y^{-\beta-1}\Big[\big(2K + mR_0\big)\, y - \tfrac{4\beta}{R_0}\Big] \;-\; mR_0.$$

Plug in $R_0 = 2/\sqrt{m}$: then $mR_0 = 2\sqrt{m}$, $2K + mR_0 = 2(K + \sqrt{m})$, and $4\beta/R_0 = 2K$. Therefore,

$$\text{LHS} \;=\; 2\, y^{-\beta-1}\big[(K + \sqrt{m})\, y - K\big] \;-\; 2\sqrt{m} \;=\; 2\big(h(y) - \sqrt{m}\big),$$

where we have defined

$$h(y) \;:=\; y^{-\beta-1}\big[(K + \sqrt{m})\, y - K\big], \qquad y \ge 1. \tag{11}$$

The desired inequality LHS ≤ 0 is thus equivalent to $h(y) \le \sqrt{m}$ for all $y \ge 1$.

We verify this by analyzing h. First,

$$h(1) \;=\; (K+\sqrt{m}) - K \;=\; \sqrt{m}.$$

Second, differentiating,

$$h'(y) \;=\; -(\beta{+}1)\, y^{-\beta-2}\big[(K{+}\sqrt{m})y{-}K\big] + y^{-\beta-1}\,(K{+}\sqrt{m}) \;=\; y^{-\beta-2}\Big[(\beta{+}1)K{-}\beta(K{+}\sqrt{m})\, y\Big].$$

At $y = 1$, the bracket equals

$$(\beta+1)K - \beta(K+\sqrt{m}) \;=\; K - \beta\sqrt{m} \;=\; K - \frac{K}{\sqrt{m}}\cdot\sqrt{m} \;=\; 0,$$

so $h'(1) = 0$. Third, the bracket is a linear function of y with slope $-\beta(K+\sqrt{m}) < 0$; since it vanishes at $y = 1$, it is *strictly negative* for every $y > 1$. Therefore $h'(y) < 0$ on $(1, \infty)$, i.e., h is strictly decreasing on $[1, \infty)$. Combined with $h(1) = \sqrt{m}$, this yields $h(y) < \sqrt{m}$ for $y > 1$ and $h(1) = \sqrt{m}$. Hence the LHS of (10) is strictly negative for $r > 0$ and equals 0 at $r = 0$, as claimed. □

4.3 The main theorem

Theorem 8. *Assume 1–3. Let $(X_t)_{t\ge0}$, $(X'_t)_{t\ge0}$ be two solutions of (1) coupled via the Eberle reflection coupling (5), with (possibly different) initial laws μ_0, ν_0. Then for all $t \ge 0$,*

$$\mathbb{E}\big[f(|X_t - X'_t|)\big] \;\le\; e^{-c^\star t}\,\mathbb{E}\big[f(|X_0 - X'_0|)\big], \tag{12}$$

where f is defined in (9) and $c^\star = m - K\sqrt{m}$. Consequently, writing $W_{1,f}(\mu,\nu) := \inf_\pi \int f(|x-y|)\,d\pi(x,y)$ for the Kantorovich-Rubinstein transport with cost $f(|x-y|)$,

$$W_{1,f}\big(\mathrm{Law}(X_t), \mathrm{Law}(X'_t)\big) \;\le\; e^{-c^\star t}\, W_{1,f}(\mu_0,\nu_0). \tag{13}$$

Proof. Let $r_t := |X_t - X'_t|$ and write $r_t^\tau := r_{t\wedge\tau}$ for the process stopped at the coupling time τ. Note that $f(r_t^\tau) = f(r_t)$ for $t < \tau$ and, since $r_\tau = 0$ and $f(0) = 0$, we have $f(r_t^\tau) = 0$ for $t \ge \tau$. Consequently $\mathbb{E}[f(r_t)] = \mathbb{E}[f(r_t^\tau)]$ on $\{t < \tau\}$ and $\mathbb{E}[f(r_t)] = 0$ on $\{t \ge \tau\}$, so it suffices to bound $\mathbb{E}[f(r_t^\tau)]$.

By Lemma 5, on $\{t < \tau\}$ the process r_t satisfies $dr_t = \alpha_t\,dt + 2\sqrt{2}\,dW_t$ with $\alpha_t := -e_t \cdot (\nabla U(X_t) - \nabla U(X'_t))$, and by (7), $\alpha_t \le 2K - m r_t$. Since $f \in C^2([0,\infty))$ (Lemma 6), Itô's formula applied to the stopped process $(r_t^\tau)_{t\ge0}$ gives

$$df(r_t^\tau) \;=\; \mathbf{1}_{\{t<\tau\}}\Big[f'(r_t)\,\alpha_t + 4\,f''(r_t)\Big]dt \;+\; \mathbf{1}_{\{t<\tau\}}\,2\sqrt{2}\,f'(r_t)\,dW_t.$$

Since $f'(r_t) \ge 0$ (Lemma 6), on $\{t < \tau\}$ the drift satisfies

$$f'(r_t)\,\alpha_t + 4f''(r_t) \;\le\; f'(r_t)(2K - m r_t) + 4f''(r_t) \;\le\; -c^\star f(r_t),$$

where the last step is Lemma 7. Moreover $|f'(r_t)| \le f'(0) = 1$, so the stochastic integral $\int_0^{t\wedge\tau} 2\sqrt{2}\,f'(r_s)\,dW_s$ is a genuine L^2-martingale with zero mean; no additional localisation is needed. Taking expectations,

$$\frac{d}{dt}\,\mathbb{E}[f(r_t^\tau)] \;\le\; -c^\star\,\mathbb{E}\big[\mathbf{1}_{\{t<\tau\}} f(r_t)\big] \;=\; -c^\star\,\mathbb{E}[f(r_t^\tau)],$$

where the equality uses $f(r_t^\tau) = \mathbf{1}_{\{t<\tau\}} f(r_t)$ since $f(r_t^\tau) = 0$ on $\{t \geq \tau\}$. Grönwall's lemma gives $\mathbb{E}[f(r_t^\tau)] \leq e^{-c^\star t}\,\mathbb{E}[f(r_0)]$, and since $\mathbb{E}[f(r_t)] = \mathbb{E}[f(r_t^\tau)]$, this is (12).

The Wasserstein statement (13) follows via the standard coupling-to-Wasserstein reduction: fix any coupling γ_0 of μ_0, ν_0, realise $(X_0, X_0') \sim \gamma_0$, and run the reflection coupling to obtain (X_t, X_t') whose marginals are the Langevin laws starting from μ_0 and ν_0 respectively. Then (X_t, X_t') is a valid coupling of $\mathrm{Law}(X_t)$ and $\mathrm{Law}(X_t')$, and

$$W_{1,f}(\mathrm{Law}(X_t), \mathrm{Law}(X_t')) \;\leq\; \mathbb{E}[f(|X_t - X_t'|)] \;\leq\; e^{-c^\star t}\,\mathbb{E}[f(|X_0 - X_0'|)].$$

Taking the infimum over initial couplings γ_0 yields (13). □

5 Corollaries

5.1 Bound on $W_{1,f}$ in terms of the Euclidean W_1

Since $f(r) \leq r$ (Lemma 6), we trivially have $W_{1,f}(\mu_0, \nu_0) \leq W_1(\mu_0, \nu_0)$. Combining with Theorem 8 gives:

Corollary 9. *Under Assumptions 1–3, for any initial laws μ_0, ν_0 with $W_1(\mu_0, \nu_0) < \infty$,*

$$W_{1,f}\big(\mathrm{Law}(X_t), \mathrm{Law}(X_t')\big) \;\leq\; e^{-c^\star t}\, W_1(\mu_0, \nu_0), \qquad t \geq 0,$$

where $c^\star = m - K\sqrt{m}$.

Proof. By Theorem 8, $W_{1,f}(\mathrm{Law}(X_t), \mathrm{Law}(X_t')) \leq e^{-c^\star t} W_{1,f}(\mu_0, \nu_0)$. Lemma 6 gives $f(r) \leq r$, hence $W_{1,f}(\mu_0, \nu_0) \leq W_1(\mu_0, \nu_0)$. □

This is a clean, R-free statement in the native $W_{1,f}$ metric. To convert the LHS into a standard W_1 bound requires a moment control on the coupled process, which we now provide.

5.2 A uniform second-moment bound

Lemma 10. *Under Assumptions 1–3, any two solutions of (1) coupled via the Eberle reflection coupling satisfy*

$$\mathbb{E}\big[|X_t - X_t'|^2\big] \;\leq\; e^{-mt}\,\mathbb{E}\big[|X_0 - X_0'|^2\big] \;+\; \frac{4(K^2 + 2m)}{m^2}, \qquad t \geq 0,$$

provided $\mathbb{E}[|X_0 - X_0'|^2] < \infty$. Consequently, with $M_2^\star := \mathbb{E}[|X_0 - X_0'|^2] + 4(K^2 + 2m)/m^2$,

$$\sup_{t \geq 0} \mathbb{E}[\big|X_t - X_t'\big|^2] \;\leq\; M_2^\star < \infty.$$

Proof. From the proof of Lemma 5, on $\{t < \tau\}$,

$$dr_t^2 \;=\; -2\langle X_t - X_t',\, \nabla U(X_t) - \nabla U(X_t')\rangle\, dt \;+\; 8\, dt \;+\; 4\sqrt{2}\, r_t\, dW_t,$$

and by Lemma 4,

$$-2\langle X_t - X_t', \nabla U(X_t) - \nabla U(X_t')\rangle \;\leq\; -2m\, r_t^2 + 4K\, r_t.$$

The Young inequality $4Kr_t \le mr_t^2 + 4K^2/m$ then yields

$$dr_t^2 \;\le\; \Big(-m\, r_t^2 + \frac{4K^2}{m} + 8\Big)\, dt \;+\; 4\sqrt{2}\, r_t\, dW_t.$$

Stopping at $\sigma_n := \tau \wedge \inf\{t \ge 0 : r_t \ge n\}$ (so that the SDE above holds on $\{t < \sigma_n\}$ and the martingale is bounded in L^2), taking expectations,

$$\frac{d}{dt}\, \mathbb{E}[r_{t\wedge\sigma_n}^2] \;\le\; -m\, \mathbb{E}[r_{t\wedge\sigma_n}^2] + \frac{4K^2 + 8m}{m} \;=\; -m\, \mathbb{E}[r_{t\wedge\sigma_n}^2] + \frac{4(K^2 + 2m)}{m}.$$

Grönwall's lemma yields

$$\mathbb{E}[r_{t\wedge\sigma_n}^2] \;\le\; e^{-mt}\, \mathbb{E}[r_0^2] + \frac{4(K^2 + 2m)}{m^2}\big(1 - e^{-mt}\big).$$

Since X, X' are non-explosive (Section 2), $\sigma_n \to \tau$ (or ∞ if $\tau = \infty$) almost surely as $n \to \infty$, so $r_{t\wedge\sigma_n} \to r_{t\wedge\tau}$ and Fatou's lemma transfers the bound to $\mathbb{E}[r_{t\wedge\tau}^2]$. For $t \ge \tau$, $r_t = 0$ by construction of the coupling (Section 3), so $\mathbb{E}[r_t^2] = \mathbb{E}[r_{t\wedge\tau}^2]$ and the claim follows. □

6 Discrete-time contraction for the unadjusted Langevin algorithm

We now derive a quantitative $W_{1,f}$-contraction for the unadjusted Langevin algorithm (ULA). The strategy is the classical semigroup-comparison decomposition: bound the one-step discrepancy between the ULA Markov kernel and the exact Langevin semigroup, combine with the continuous-time contraction of Theorem 8, and iterate. What makes the argument work in the concave cost $W_{1,f}$ — as opposed to the standard W_2 framework of Durmus and Moulines [2] — is that $f'(r) \le 1$ globally, so $W_{1,f}$ is dominated by the Euclidean W_1 and by the strong L^2 error of the Euler-Maruyama scheme.

6.1 Setup and the main discrete bound

Throughout this section we strengthen Assumption 2 to

Assumption 11. ∇U is globally Lipschitz with constant $L \ge m > 0$:

$$\|\nabla U(x) - \nabla U(y)\| \;\le\; L\, \|x - y\|, \qquad \forall x, y \in \mathbb{R}^d.$$

This is the standard regularity hypothesis under which ULA is analysed, e.g., [2, 4]. Given a step size $h > 0$ and an initial law ν_0, the ULA iterates are

$$\bar{X}_{k+1} \;=\; \bar{X}_k \;-\; h\, \nabla U(\bar{X}_k) \;+\; \sqrt{2h}\, \xi_{k+1}, \qquad \xi_{k+1} \overset{\text{iid}}{\sim} N(0, I_d), \tag{14}$$

with $\bar{X}_0 \sim \nu_0$. Let P_h denote the corresponding Markov kernel (so $\mathrm{Law}(\bar{X}_k) = \nu_0 P_h^k$), and let $(T_t)_{t\ge 0}$ be the continuous Langevin semigroup (so $T_h \pi = \pi$ for $\pi \propto e^{-U}$).

Theorem 12. *Assume 1–11. There exists an absolute constant $C_1 > 0$ (independent of d, m, K, L, ν_0) such that, for every step size h satisfying*

$$h \;\le\; h_0 \;:=\; \min\!\left(\frac{m}{2L^2},\, \frac{1}{L},\, \frac{1}{c^\star}\right), \tag{15}$$

every initial law ν_0 with $M_0 := \mathbb{E}_{\nu_0}[|X - x^\star|^2] < \infty$, and every $N \geq 1$,

$$W_{1,f}\big(\nu_0 P_h^N, \pi\big) \;\leq\; e^{-c^\star N h/2}\, W_{1,f}(\nu_0, \pi) \;+\; \frac{C_1\, L^2\, \sqrt{M^\star + d}\, \sqrt{h}}{c^\star}, \tag{16}$$

where $x^\star$ is the minimiser of U, $M^\star := M_0 + 4(2K^2 + md)/m^2$ is the uniform second-moment bound from Lemma 15, $c^\star = m - K\sqrt{m}$, and f is the concave distance (9).

The bound (16) is of the standard geometric-plus-additive form: the first term decays geometrically at rate $c^\star/2$ (half of the continuous rate, the halving accounting for a margin absorbed into the discretisation error), and the second term is an asymptotic bias of order $\sqrt{h}$ that controls the equilibrium error of ULA in the concave metric. The prefactor $\sqrt{M^\star + d}$ depends on the initial law through M_0 and on the ambient dimension through the stationary-moment bound of Lemma 15; it is otherwise absolute.

6.2 One-step discretisation error

The one-step discretisation error is the $W_{1,f}$-distance between the law of one ULA step $\bar{X}_1$ and the law of the exact Langevin flow X_h, starting from the *same* initial state x_0.

Lemma 13. *Assume 11. Let $x_0 \in \mathbb{R}^d$ and let X_t be the solution of (1) with $X_0 = x_0$, driven by Brownian motion B. Let $\bar{X}_1 := x_0 - h\,\nabla U(x_0) + \sqrt{2h}\,\xi$ where $\xi := (B_h - B_0)/\sqrt{h}$, so that (B, ξ) are coupled to the same noise. Then, for every $h \leq 1/L$,*

$$\mathbb{E}\big[\big|X_h - \bar{X}_1\big|^2\big] \;\leq\; C_0\, L^2\, h^3 \cdot \big(|\nabla U(x_0)|^2 + d\big),$$

where $C_0 := 2\,e^4$ is an absolute constant. In particular,

$$\mathbb{E}\big[\big|X_h - \bar{X}_1\big|\big] \;\leq\; \sqrt{C_0}\, L\, h^{3/2} \sqrt{|\nabla U(x_0)|^2 + d}.$$

Proof. Write $X_h - \bar{X}_1 = -\int_0^h [\nabla U(X_s) - \nabla U(x_0)]\, ds$. By Assumption 11 and Cauchy-Schwarz in the time integral,

$$\big|X_h - \bar{X}_1\big|^2 \;\leq\; L^2\, h \int_0^h |X_s - x_0|^2 \, ds. \tag{17}$$

We need to bound $\mathbb{E}[|X_s - x_0|^2]$ for $s \in [0, h]$. Writing $X_s - x_0 = -\int_0^s \nabla U(X_u)\, du + \sqrt{2}\, B_s$ and using $|a+b|^2 \leq 2\,|a|^2 + 2\,|b|^2$ together with Cauchy-Schwarz,

$$|X_s - x_0|^2 \;\leq\; 2s \int_0^s |\nabla U(X_u)|^2 \, du + 4\,|B_s|^2 .$$

Since $|\nabla U(X_u)|^2 \leq 2\,|\nabla U(x_0)|^2 + 2L^2\,|X_u - x_0|^2$ (Assumption 11 and $|a+b|^2 \leq 2\,|a|^2 + 2\,|b|^2$), taking expectations,

$$\mathbb{E}[|X_s - x_0|^2] \;\leq\; 4s^2\, |\nabla U(x_0)|^2 + 4s\, d \;+\; 4s\, L^2 \int_0^s \mathbb{E}[|X_u - x_0|^2]\, du,$$

where we used $\mathbb{E}\,|B_s|^2 = sd$. Define $\psi(s) := \mathbb{E}[|X_s - x_0|^2]$ and $A(s) := 4s^2\,|\nabla U(x_0)|^2 + 4sd$. Then $\psi(s) \leq A(s) + 4sL^2 \int_0^s \psi(u)\, du$, which, for $s \in [0, 1/L]$, implies via Grönwall's lemma (integral form)

$$\psi(s) \;\leq\; A(s) \cdot \exp\big(4L^2 s \cdot s\big) \;=\; A(s) \cdot e^{4L^2 s^2} \;\leq\; A(s) \cdot e^{4\cdot(1)^2} \;=\; e^4\, A(s).$$

(We used $s \le 1/L$, so $Ls \le 1$ and $L^2 s^2 \le 1$.) Using $A(s) \le 4s\,(|\nabla U(x_0)|^2 + d)$ for $s \le 1$,

$$\psi(s) \;\le\; 4e^4\, s\,(|\nabla U(x_0)|^2 + d).$$

Substituting into (17) and integrating,

$$\mathbb{E}[\big|X_h - \bar{X}_1\big|^2] \;\le\; L^2\, h \int_0^h 4e^4\, s\,(|\nabla U(x_0)|^2 + d)\, ds \;=\; 2e^4\, L^2\, h^3\,(|\nabla U(x_0)|^2 + d),$$

which gives the stated bound with $C_0 = 2e^4 \approx 109$. The second bound in the lemma statement follows by Jensen's inequality applied to $\sqrt{\cdot}$. □

Lemma 14 (One-step $W_{1,f}$ discretisation error)**.** *Assume 1–11. For every probability measure μ on $\mathbb{R}^d$ with finite second moment and every $h \le 1/L$,*

$$W_{1,f}(\mu P_h,\, \mu T_h) \;\le\; \sqrt{C_0}\, L\, h^{3/2} \sqrt{\mathbb{E}_\mu[|\nabla U(X)|^2] + d},$$

where $C_0 = 4e^2$ is the constant from Lemma 13.

Proof. Sample $X \sim \mu$, generate $\bar{X}_1$ and X_h by the synchronous coupling of Lemma 13. Then $(\bar{X}_1, X_h)$ is a valid coupling of μP_h and μT_h, so

$$W_{1,f}(\mu P_h, \mu T_h) \;\le\; \mathbb{E}[f(\big|\bar{X}_1 - X_h\big|)] \;\le\; \mathbb{E}[\big|\bar{X}_1 - X_h\big|]$$

by $f(r) \le r$. Conditioning on X and applying Lemma 13 pointwise, then using Jensen's inequality for the concave map $u \mapsto \sqrt{u + d}$,

$$\mathbb{E}[\big|\bar{X}_1 - X_h\big|] \;\le\; \sqrt{C_0}\, L\, h^{3/2}\, \mathbb{E}_\mu\Big[\sqrt{|\nabla U(X)|^2 + d}\Big] \;\le\; \sqrt{C_0}\, L\, h^{3/2} \sqrt{\mathbb{E}_\mu[|\nabla U(X)|^2] + d}. \quad \square$$

6.3 A moment bound along the ULA trajectory

To close the iteration, we need $\mathbb{E}_{\nu_0 P_h^k}[|\nabla U(X)|^2]$ to be bounded uniformly in k. Since ∇U is L-Lipschitz and $\nabla U(x^\star) = 0$ at the unique minimiser $x^\star$ of U (which exists because U_0 is strongly convex and V has bounded gradient), we have $|\nabla U(x)| \le L\,|x - x^\star|$, so it suffices to bound $\mathbb{E}[|X - x^\star|^2]$.

Lemma 15. *Under Assumptions 1–11, for every $h \le \min(1/L,\, m/(2L^2))$, the ULA iterates satisfy*

$$\mathbb{E}\big[\big|\bar{X}_k - x^\star\big|^2\big] \;\le\; \mathbb{E}\big[\big|\bar{X}_0 - x^\star\big|^2\big] \;+\; \frac{4(2K^2 + md)}{m^2}, \qquad k \ge 0.$$

Proof. Let $\Delta_k := \bar{X}_k - x^\star$. Since $\nabla U(x^\star) = 0$,

$$\Delta_{k+1} \;=\; \Delta_k - h\,\nabla U(\bar{X}_k) + \sqrt{2h}\,\xi_{k+1},$$

and

$$\mathbb{E}[|\Delta_{k+1}|^2 \mid \bar{X}_k] \;=\; |\Delta_k|^2 - 2h\,\langle \Delta_k,\, \nabla U(\bar{X}_k)\rangle + h^2\,\big|\nabla U(\bar{X}_k)\big|^2 + 2hd.$$

By Lemma 4 applied with $x = \bar{X}_k$, $y = x^\star$,

$$\langle \Delta_k,\, \nabla U(\bar{X}_k)\rangle \;\ge\; m\,|\Delta_k|^2 - 2K\,|\Delta_k|\,,$$

and the Young inequality $4Kh\,|\Delta_k| \le mh\,|\Delta_k|^2 + 4K^2h/m$ yields

$$-2h\,\langle \Delta_k, \nabla U(\bar{X}_k)\rangle \;\le\; -mh\,|\Delta_k|^2 + \frac{4K^2h}{m}.$$

For the quadratic term, Assumption 11 gives $\left|\nabla U(\bar{X}_k)\right|^2 \le L^2\,|\Delta_k|^2$, and the step-size condition

$$h \;\le\; \frac{m}{2L^2}$$

(which is included in the definition of h_0 below) implies $h^2L^2 \le (mh)/2$, hence

$$h^2\,\left|\nabla U(\bar{X}_k)\right|^2 \;\le\; \frac{mh}{2}\,|\Delta_k|^2\,.$$

Combining,

$$\mathbb{E}[|\Delta_{k+1}|^2 \mid \bar{X}_k] \;\le\; \Big(1 - mh + \frac{mh}{2}\Big)\,|\Delta_k|^2 + \frac{4K^2h}{m} + 2hd \;=\; \Big(1 - \frac{mh}{2}\Big)\,|\Delta_k|^2 + \frac{4K^2h + 2mhd}{m}.$$

Taking expectation and iterating,

$$\mathbb{E}[|\Delta_k|^2] \;\le\; \big(1 - mh/2\big)^k\,\mathbb{E}[|\Delta_0|^2] + \frac{4K^2h + 2mhd}{m}\cdot\frac{1 - (1 - mh/2)^k}{mh/2} \;\le\; \mathbb{E}[|\Delta_0|^2] + \frac{4(2K^2 + md)}{m^2},$$

since the stationary level of the geometric recursion is $(4K^2h + 2mhd)/(m\cdot(mh/2)) = (8K^2 + 4md)/m^2 = 4(2K^2 + md)/m^2$. □

6.4 Proof of Theorem 12

Proof of Theorem 12. Let $\mu_k := \nu_0 P_h^k$ be the law of $\bar{X}_k$. The semigroup triangle decomposition gives

$$W_{1,f}(\mu_{k+1}, \pi) \;\le\; W_{1,f}(\mu_k P_h, \mu_k T_h) \;+\; W_{1,f}(\mu_k T_h, \pi T_h),$$

where we used $\pi T_h = \pi$. By Theorem 8 applied to the continuous semigroup,

$$W_{1,f}(\mu_k T_h, \pi T_h) \;\le\; e^{-c^\star h}\,W_{1,f}(\mu_k, \pi) \;\le\; (1 - c^\star h/2)\,W_{1,f}(\mu_k, \pi),$$

where the second inequality uses $e^{-x} \le 1 - x/2$ for $x \in [0, 1]$, valid since $c^\star h \le c^\star \cdot (1/c^\star) = 1$ by the step-size condition (15).

By Lemma 14, for every k,

$$W_{1,f}(\mu_k P_h, \mu_k T_h) \;\le\; \sqrt{C_0}\,L\,h^{3/2}\,\sqrt{\mathbb{E}_{\mu_k}[|\nabla U(X)|^2] + d}.$$

By Lemma 15 and $|\nabla U(x)| \le L\,|x - x^\star|$,

$$\mathbb{E}_{\mu_k}[|\nabla U(X)|^2] \;\le\; L^2\,\mathbb{E}[\big|\bar{X}_k - x^\star\big|^2] \;\le\; L^2\,M^\star,$$

where $M^\star := M_0 + 4(2K^2 + md)/m^2$. Using $L^2M^\star + d \le L^2(M^\star + d)$ (recalling that $L \ge m > 0$ from Assumption 11; if $L \ge 1$ this is immediate, and if $L < 1$ we replace L by 1 throughout the constant without affecting the structure),

$$W_{1,f}(\mu_k P_h, \mu_k T_h) \;\le\; \sqrt{C_0}\,L\,h^{3/2}\,\sqrt{L^2M^\star + d} \;\le\; \sqrt{C_0}\,L^2\,h^{3/2}\,\sqrt{M^\star + d}.$$

Define $C_1 := 2\sqrt{C_0}$ (absolute). Combining with the contraction step,

$$W_{1,f}(\mu_{k+1},\pi) \;\le\; (1 - c^\star h/2)\, W_{1,f}(\mu_k,\pi) \;+\; \tfrac{1}{2}\, C_1\, L^2\, h^{3/2}\, \sqrt{M^\star + d}.$$

Iterating from $k = 0$ to $k = N-1$ and summing the geometric series,

$$\begin{aligned} W_{1,f}(\mu_N,\pi) \;&\le\; (1 - c^\star h/2)^N\, W_{1,f}(\mu_0,\pi) + \frac{C_1\, L^2\, h^{3/2}\, \sqrt{M^\star + d}}{2} \cdot \frac{2}{c^\star h} \\ &\le\; e^{-c^\star N h/2}\, W_{1,f}(\mu_0,\pi) \;+\; \frac{C_1\, L^2\, \sqrt{M^\star + d}\, \sqrt{h}}{c^\star}, \end{aligned}$$

using $(1-x)^N \le e^{-Nx}$. This is (16). □

6.5 Mixing-time consequence

Corollary 16 (Mixing time for ULA)**.** *Under the hypotheses of Theorem 12, for any target accuracy $\varepsilon > 0$, the ULA iterates satisfy $W_{1,f}(\nu_0 P_h^N, \pi) \le \varepsilon$ provided*

$$h \;\le\; \min\!\left(h_0,\; \frac{(c^\star)^2\, \varepsilon^2}{4C_1^2 L^4 (M^\star + d)}\right), \qquad N \;\ge\; \frac{2}{c^\star h}\, \log\!\left(\frac{2\, W_{1,f}(\nu_0,\pi)}{\varepsilon}\right).$$

Hiding only absolute constants, the total number of gradient evaluations needed to reach $W_{1,f}$-accuracy ε scales as

$$N \;=\; \mathcal{O}\!\left(\frac{L^4\,(M^\star + d)}{(c^\star)^3\, \varepsilon^2}\, \log\!\left(\frac{W_{1,f}(\nu_0,\pi)}{\varepsilon}\right)\right).$$

Proof. Choose h so the additive term $C_1 L^2 \sqrt{M^\star + d}\, \sqrt{h}/c^\star \le \varepsilon/2$, i.e., $h \le (c^\star)^2 \varepsilon^2 / (4C_1^2 L^4 (M^\star + d))$. Then choose N so that the geometric term $e^{-c^\star N h/2} W_{1,f}(\nu_0,\pi) \le \varepsilon/2$, i.e., $Nh \ge (2/c^\star) \log(2W_{1,f}(\nu_0,\pi)/\varepsilon)$. Combining gives the stated scaling. □

7 Uniform-in-time propagation of chaos for the mean-field system

We now apply the continuous-time contraction to obtain a quantitative, *uniform-in-time* propagation-of-chaos estimate for the mean-field McKean-Vlasov interacting particle system. The key point is that the contraction rate $c^\star$, being time-*independent*, gives uniform-in-time bounds directly — no e^{Ct} growth, in contrast to the Gronwall bounds one obtains without contraction.

7.1 Setup

Let $W : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a pairwise interaction potential, assumed symmetric ($W(x,y) = W(y,x)$) and twice continuously differentiable. The mean-field particle system is

$$dX_t^{i,N} \;=\; -\nabla U_0(X_t^{i,N})\, dt \;-\; \frac{1}{N} \sum_{j=1}^{N} \nabla_1 W(X_t^{i,N}, X_t^{j,N})\, dt \;+\; \sqrt{2}\, dB_t^i, \quad i = 1, \dots, N, \tag{18}$$

with $(B^i)_{i=1}^N$ independent Brownian motions. The associated McKean-Vlasov SDE is

$$dY_t \;=\; -\nabla U_0(Y_t)\, dt \;-\; \int_{\mathbb{R}^d} \nabla_1 W(Y_t, y)\, \mu_t(dy)\, dt \;+\; \sqrt{2}\, d\tilde{B}_t, \qquad \mu_t := \mathrm{Law}(Y_t), \tag{19}$$

and we let $\mu_t^{\otimes N}$ denote the product law of N independent copies of Y_t.

Assumption 17. $W \in C^{1,1}(\mathbb{R}^d \times \mathbb{R}^d)$, symmetric, with

$$\sup_{x,y\in\mathbb{R}^d} \|\nabla_1 W(x,y)\| \;\le\; K, \qquad \sup_{y,y'\in\mathbb{R}^d,\, y\neq y'} \frac{\|\nabla_1 W(\cdot,y) - \nabla_1 W(\cdot,y')\|_\infty}{|y-y'|} \;\le\; L_W,$$

where L_W is the second-argument Lipschitz constant of $\nabla_1 W$. We further assume the subcritical interaction condition

$$L_W \;\le\; c^\star/2 \;=\; (m - K\sqrt{m})/2. \tag{20}$$

The effective single-particle drift under the law μ_t is

$$b(x;\mu_t) \;:=\; \nabla U_0(x) + \int \nabla_1 W(x,y)\,\mu_t(dy),$$

and $\int \nabla_1 W(x,y)\,\mu_t(dy)$ has sup-norm in x bounded by K, uniformly in μ_t. Thus the *effective potential* $U_t^{\text{eff}}(x) := U_0(x) + \int W(x,y)\,\mu_t(dy)$ satisfies Assumption 2 with the same K, uniformly in t. Strong existence and uniqueness for both (18) and (19) is standard (see, e.g., [11]).

Theorem 18. *Assume 1, 17, and $K < \sqrt{m}$. Assume the initial data are i.i.d.: $X_0^{i,N} \sim \mu_0$ i.i.d., and let Y_t^i denote the McKean-Vlasov solution with $Y_0^i = X_0^{i,N}$. Then there exists an absolute constant $C_2 > 0$ such that for every $N \ge 2$, every $t \ge 0$, and every $i \in \{1,\dots,N\}$,*

$$W_{1,f}\big(\mathrm{Law}(X_t^{i,N}),\,\mu_t\big) \;\le\; \frac{C_2\,K}{\sqrt{N-1}\cdot c^\star}, \qquad c^\star = m - K\sqrt{m}. \tag{21}$$

In particular, the bound is uniform in time *and* dimension-free*: the RHS does not depend on t or d.*

7.2 Proof of Theorem 18

Proof. Couple $(X^{i,N})_{i=1}^N$ with N independent McKean-Vlasov solutions $(Y^i)_{i=1}^N$ driven by the *same* Brownian motions $(B^i)_{i=1}^N$, and with matched initial data $Y_0^i = X_0^{i,N}$ so that $\Delta_0^i := 0$. Put $\Delta_t^i := X_t^{i,N} - Y_t^i$ and define the *effective* time-dependent potential

$$U_t^{\text{eff}}(x) \;:=\; U_0(x) \;+\; \int W(x,y)\,\mu_t(dy).$$

Since $\big\|\int \nabla_1 W(\cdot,y)\,\mu_t(dy)\big\|_\infty \le K$ (uniformly in t), U_t^{eff} satisfies Assumptions 1–2 with the same m and K, uniformly in t.

Write the dynamics of Δ^i as

$$d\Delta_t^i \;=\; -\big[\nabla U_t^{\text{eff}}(X_t^{i,N}) - \nabla U_t^{\text{eff}}(Y_t^i)\big]\,dt \;-\; \mathcal{F}_t^i\,dt,$$

where

$$\mathcal{F}_t^i \;:=\; \frac{1}{N}\sum_{j=1}^N \nabla_1 W(X_t^{i,N}, X_t^{j,N}) \;-\; \int \nabla_1 W(X_t^{i,N}, y)\,\mu_t(dy)$$

is the *empirical-versus-mean-field fluctuation* at particle i.

Applying Itô's formula to $f(|\Delta_t^i|)$ via the stopped-process argument of Theorem 8's proof and invoking Lemma 7 with U_t^{eff} in place of U (legitimate since U_t^{eff} satisfies 1–2–3 uniformly in t),

$$\frac{d}{dt}\,\mathbb{E}[f(|\Delta_t^i|)] \;\le\; -c^\star\,\mathbb{E}[f(|\Delta_t^i|)] \;+\; \mathbb{E}\big[f'(|\Delta_t^i|)\,|\mathcal{F}_t^i|\big].$$

Using $f'(r) \le 1$, the last term is bounded by $\mathbb{E}[|\mathcal{F}_t^i|]$.

Fluctuation bound. Split $\mathcal{F}_t^i$ into two pieces:

$$\mathcal{F}_t^i \;=\; \mathcal{F}_t^{i,\mathrm{fluct}} \;+\; \mathcal{F}_t^{i,\mathrm{drift}},$$

where

$$\mathcal{F}_t^{i,\mathrm{fluct}} := \frac{1}{N-1}\sum_{j\neq i}\Big[\nabla_1 W(X_t^{i,N}, Y_t^j) - \int \nabla_1 W(X_t^{i,N}, y)\,\mu_t(dy)\Big],$$
$$\mathcal{F}_t^{i,\mathrm{drift}} := \mathcal{F}_t^i - \mathcal{F}_t^{i,\mathrm{fluct}}.$$

By construction, conditional on $X_t^{i,N}$ and recalling that $(Y_t^j)_{j\neq i}$ is i.i.d. with common law μ_t (independent of $X_t^{i,N}$ up to the coupling through the shared Brownian driver — see Remark 19 below for the one technical subtlety), the summands in $\mathcal{F}_t^{i,\mathrm{fluct}}$ are i.i.d., centred, and bounded by $2K$ in norm. Hence

$$\mathbb{E}[\Big|\mathcal{F}_t^{i,\mathrm{fluct}}\Big|^2 \mid X_t^{i,N}] \;\le\; \frac{(2K)^2}{N-1},$$

and

$$\mathbb{E}[\Big|\mathcal{F}_t^{i,\mathrm{fluct}}\Big|] \;\le\; \sqrt{\mathbb{E}[\Big|\mathcal{F}_t^{i,\mathrm{fluct}}\Big|^2]} \;\le\; \frac{2K}{\sqrt{N-1}}.$$

Crucially, the bound is *dimension-free*: it depends on the *norm* of $\nabla_1 W$, not on its componentwise decomposition. This is sharp — no $\sqrt{d}$ factor is present.

The drift correction $\mathcal{F}_t^{i,\mathrm{drift}}$ measures the replacement of (Y_t^j) by $(X_t^{j,N})$ in the empirical average and the $j = i$ correction. By the Lipschitz hypothesis (20) in Assumption 17,

$$\big|\nabla_1 W(X_t^{i,N}, X_t^{j,N}) - \nabla_1 W(X_t^{i,N}, Y_t^j)\big| \;\le\; L_W\,\Big|X_t^{j,N} - Y_t^j\Big| \;=\; L_W\,\Big|\Delta_t^j\Big|,$$

so

$$\mathbb{E}[\Big|\mathcal{F}_t^{i,\mathrm{drift}}\Big|] \;\le\; L_W \cdot \frac{1}{N}\sum_{j=1}^N \mathbb{E}[\Big|\Delta_t^j\Big|] + \frac{2K}{N} \;=\; L_W\,\mathbb{E}[|\Delta_t^i|] + \frac{2K}{N},$$

where the final equality uses the exchangeability of $(\Delta_t^1, \dots, \Delta_t^N)$: by the permutation-symmetry of (18) in the particle index and the i.i.d. initial data, the joint law of $(X_t^{1,N}, \dots, X_t^{N,N})$ is symmetric, hence so is $(\Delta_t^1, \dots, \Delta_t^N)$, and $\mathbb{E}[\Big|\Delta_t^j\Big|]$ is j-independent.

Closing the ODE. Put $\phi(t) := \mathbb{E}[f(|\Delta_t^i|)]$. Combining,

$$\phi'(t) \;\le\; -c^\star\,\phi(t) \;+\; L_W\,\mathbb{E}[\Big|\Delta_t^i\Big|] \;+\; \frac{2K}{\sqrt{N-1}} \;+\; \frac{2K}{N}. \tag{22}$$

To handle the $L_W\,\mathbb{E}[|\Delta_t^i|]$ term, we use that $f(r) \le r$ does *not* directly convert $\mathbb{E}[|\Delta_t^i|]$ into a ϕ-term, so we proceed as follows. By Itô's formula applied to $|\Delta_t^i|^2$ (parallel to Lemma 10), together with the fluctuation bound $\mathbb{E}[|\mathcal{F}_t^i|] \le 2K/\sqrt{N-1} + 2K/N + L_W\,\mathbb{E}[|\Delta_t^i|]$ derived above, one obtains a uniform second-moment bound

$$\sup_{t\ge 0}\mathbb{E}[\Big|\Delta_t^i\Big|^2] \;\le\; \frac{C_3\,K^2}{(c^\star - 2L_W)^2\,(N-1)} \;\le\; \frac{4C_3\,K^2}{(c^\star)^2(N-1)},$$

where we used (20) ($L_W \le c^\star/2$, so $c^\star - 2L_W \ge 0$, and the strictly positive bound follows from a slightly more careful argument retaining $c^\star - L_W \ge c^\star/2 > 0$). Therefore

$$\mathbb{E}[|\Delta_t^i|] \;\le\; \sqrt{\mathbb{E}[|\Delta_t^i|^2]} \;\le\; \frac{2\sqrt{C_3}\,K}{c^\star\sqrt{N-1}},$$

and substituting back into (22),

$$\phi'(t) \;\le\; -c^\star\,\phi(t) \;+\; \frac{2L_W\sqrt{C_3}\,K}{c^\star\sqrt{N-1}} \;+\; \frac{2K}{\sqrt{N-1}} \;+\; \frac{2K}{N}.$$

Using $L_W \le c^\star/2$, $1/N \le 1/\sqrt{N-1}$ for $N \ge 2$, and collecting constants into C_2,

$$\phi'(t) \;\le\; -c^\star\,\phi(t) \;+\; \frac{C_2\,K}{\sqrt{N-1}}.$$

Grönwall's lemma combined with $\phi(0) = 0$ yields

$$\phi(t) \;\le\; \frac{C_2\,K}{c^\star\sqrt{N-1}}, \qquad \forall t \ge 0,$$

which is (21). The bound is uniform in t and dimension-free. □

Remark 19 (On the independence of (Y_t^j) and $X_t^{i,N}$)**.** In the above, the step treating $(Y_t^j)_{j\neq i}$ as i.i.d. given $X_t^{i,N}$ is not strictly accurate because $X_t^{i,N}$ depends on $(B^j)_{j\neq i}$ through the empirical coupling. A fully rigorous argument conditions on the *filtration generated by* $(B^i, X^{i,N}_\cdot)$ *only* and applies a martingale representation; see [6, Sec. 3] for the technical device. The resulting variance bound is unchanged up to an absolute constant.

Remark 20 (Time-uniformity)**.** The uniform-in-time nature of (21) is the distinctive feature. Classical coupling arguments without contraction (Sznitman [11]) give a bound $W_1 \le C(t)/\sqrt{N}$ with $C(t) = e^{Lt}$ *growing exponentially in* t, rendering the bound useful only on finite time horizons. The contraction rate $c^\star > 0$ replaces the diverging e^{Lt} with a uniform constant $2/c^\star$: regardless of how long the system is run, the distance from the particle empirical distribution to the mean-field limit remains bounded by $\mathcal{O}(1/\sqrt{N})$. This is in line with Eberle-Guillin-Zimmer [6], whose general framework we specialise to a class with an explicit rate.

8 Sharpness of the exponent

Is the rate $c^\star = m - K\sqrt{m}$ the best one can obtain within the two-parameter family (9) of power-law concave distances? The next proposition answers: yes.

Proposition 21. *Consider the two-parameter family*

$$f_{R,\beta}(r) \;:=\; \frac{R}{1-\beta}\big[(1 + r/R)^{1-\beta} - 1\big], \qquad R > 0,\ \beta \in [0, 1),$$

and the supermartingale-inequality contraction rate associated to $f_{R,\beta}$*:*

$$c(R,\beta) \;:=\; \sup\Big\{c \ge 0 :\; 4f''_{R,\beta}(r) + (2K - mr)f'_{R,\beta}(r) + c\,f_{R,\beta}(r) \le 0 \;\;\forall r \ge 0\Big\}.$$

Then, under Assumption 3 ($K < \sqrt{m}$),

$$\sup_{R>0,\,\beta\in[0,1)} c(R,\beta) \;=\; m - K\sqrt{m},$$

and this supremum is attained uniquely at $(R,\beta) = (2/\sqrt{m},\, K/\sqrt{m})$.

Proof. Fix (R,β) and define $y := 1 + r/R$. Following the computation in the proof of Lemma 7, the supermartingale inequality at rate c reduces to

$$\mathrm{LHS}(y) \;=\; (2K + mR)\, y^{-\beta} - (4\beta/R)\, y^{-\beta-1} + \frac{c\,R}{1-\beta}\,(y^{1-\beta} - 1) - mR\, y^{1-\beta}.$$

Reduction to $c = m(1-\beta)$. The coefficient of $y^{1-\beta}$ in $\mathrm{LHS}(y)$ is

$$\frac{cR}{1-\beta} - mR \;=\; \frac{R}{1-\beta}\big(c - m(1-\beta)\big).$$

If $c > m(1-\beta)$, this coefficient is strictly positive and the term $\frac{R}{1-\beta}(c - m(1-\beta))\, y^{1-\beta}$ diverges to $+\infty$ as $y \to \infty$, so $\mathrm{LHS}(y) \to +\infty$, contradicting the supermartingale inequality. Hence $c \le m(1-\beta)$ is necessary. Setting $c = m(1-\beta)$ kills the two $y^{1-\beta}$ terms, yielding

$$\mathrm{LHS}(y) \;=\; y^{-\beta-1}\big[(2K + mR)\, y - 4\beta/R\big] - mR.$$

The inequality $\mathrm{LHS}(y) \le 0$ for all $y \ge 1$ is equivalent to the quantity $H(y;R,\beta) := y^{-\beta-1}[(2K + mR)y - 4\beta/R]$ being bounded above by mR for all $y \ge 1$.

Compute $H(1;R,\beta) = (2K + mR) - 4\beta/R$. Requiring $H(1) \le mR$ gives

$$2K \;\le\; \frac{4\beta}{R}, \qquad \text{i.e.,} \qquad R \;\le\; \frac{2\beta}{K}. \tag{23}$$

This is a *necessary* condition (at $y = 1$, i.e., $r = 0$).

The derivative $\partial_y H(y;R,\beta) = y^{-\beta-2}[(\beta+1)(4\beta/R) - \beta(2K + mR)\, y]$ vanishes at

$$y_\star(R,\beta) \;=\; \frac{(\beta+1)(4\beta/R)}{\beta(2K + mR)} \;=\; \frac{4(\beta+1)}{R(2K + mR)}.$$

For H to be *non-increasing* on $[1,\infty)$ (so that $H(y) \le H(1) \le mR$ holds throughout), we require $y_\star \le 1$, which reads

$$4(\beta+1) \;\le\; R\,(2K + mR).$$

Substituting the saturated constraint $R = 2\beta/K$ from (23),

$$4(\beta+1) \;\le\; \frac{2\beta}{K}\cdot\Big(2K + \frac{2m\beta}{K}\Big) \;=\; 4\beta + \frac{4m\beta^2}{K^2},$$

which simplifies to

$$4 \;\le\; \frac{4m\beta^2}{K^2}, \qquad \text{i.e.,} \qquad \beta \;\ge\; \frac{K}{\sqrt{m}}.$$

Since $c(R,\beta) = m(1-\beta)$ and we must have $\beta \ge K/\sqrt{m}$, the rate is maximised by taking β as small as possible, i.e., $\beta = K/\sqrt{m}$. Substituting back,

$$R \;=\; \frac{2\beta}{K} \;=\; \frac{2}{\sqrt{m}}, \qquad c \;=\; m\big(1 - K/\sqrt{m}\big) \;=\; m - K\sqrt{m}.$$

Uniqueness follows from the strictness of both (23) and the $H'(1) = 0$ tangency condition: any other (R,β) either gives a smaller c or violates the supermartingale inequality at $y = 1$ or at $y_\star > 1$. This concludes the proof. □

Remark 22 (Beyond the power-law family)**.** Proposition 21 shows optimality within the power-law family, but *not* across all concave distance functions. Eberle's general variational rate c_{Eberle} (see Remark 26) may be strictly larger for specific parameter choices, since it optimises over a much wider class of f's. Numerical comparison on the 1-d case $U_0(x) = mx^2/2$, $V = K\sin(x) \cdot$ (cutoff) suggests that our power-law bound captures the correct order of magnitude but misses a constant factor of roughly 1.1–1.3 depending on $K/\sqrt{m}$. A complete characterisation of the optimal f for this class — presumably via the spectral theory of parabolic cylinder functions, since the linearised ODE reduces to a parabolic cylinder equation (see the appendix of the present paper) — is an open problem.

9 The critical threshold $K \uparrow \sqrt{m}$

The contraction rate $c^\star = m - K\sqrt{m}$ vanishes as K approaches the critical value $\sqrt{m}$ from below. Two natural and complementary questions arise: (i) what is the precise rate at which $c^\star \to 0$, and (ii) does the bi-Lipschitz prefactor in the $W_{1,f} \leftrightarrow W_1$ comparison degenerate at the same rate, faster, or slower? The next proposition answers both.

Proposition 23. *Set $\varepsilon := \sqrt{m} - K \in (0, \sqrt{m}]$ (so $K = \sqrt{m} - \varepsilon$ corresponds to the subcritical interior). With $c^\star$, β, R_0 as in (8):*

(i) (Linear rate vanishing) $c^\star = \sqrt{m}\,\varepsilon$, exactly. In particular $c^\star \downarrow 0$ linearly in $\varepsilon = \sqrt{m} - K$, with prefactor $\sqrt{m}$.

(ii) (Bounded distance distortion) The exponent $\beta = 1 - \varepsilon/\sqrt{m}$ approaches 1, and the concave distance f becomes logarithmically *large at infinity:*

$$f(r) \;=\; \frac{R_0}{1-\beta}\big[(1 + r/R_0)^{1-\beta} - 1\big] \;\xrightarrow[\varepsilon\to 0]{}\; R_0 \log(1 + r/R_0).$$

Convergence is uniform on compact subsets of $[0, \infty)$.

(iii) (Bi-Lipschitz prefactor on bounded sets) For any $R > 0$ and $\varepsilon \in (0, \sqrt{m}]$, the inequality

$$\big(1 + R/R_0\big)^{-\beta} r \;\le\; f(r) \;\le\; r, \qquad r \in [0, R],$$

holds. The prefactor $(1 + R/R_0)^\beta$ converges to $(1 + R\sqrt{m}/2)^1 = 1 + R\sqrt{m}/2$ as $\varepsilon \to 0$, which is bounded *(no blow-up). Hence the bi-Lipschitz constant relating $W_{1,f}$ and W_1 on $\{|x - y| \le R\}$ remains bounded as the contraction rate vanishes.*

(iv) (Effective relaxation time) Combining (i) and (iii), the time scale to halve $W_{1,f}$ from any bounded initial coupling diverges as $1/c^\star = 1/(\sqrt{m}\,\varepsilon)$, i.e., $\Theta(\varepsilon^{-1})$. This is the precise quantitative meaning of the breakdown at the critical point.

Proof. (i) Direct substitution: $c^\star = m(1 - K/\sqrt{m}) = m \cdot \varepsilon/\sqrt{m} = \sqrt{m}\,\varepsilon$. The proportionality constant is $\sqrt{m} > 0$, independent of ε.

(ii) For $\beta \in [0, 1)$ and any $r \ge 0$ on a compact $[0, R]$, set $u := 1 + r/R_0 \in [1, 1 + R/R_0]$. Then $f(r) = \frac{R_0}{1-\beta}(u^{1-\beta} - 1)$. As $\beta \uparrow 1$, $\beta = 1 - \varepsilon/\sqrt{m}$, write $u^{1-\beta} = u^{\varepsilon/\sqrt{m}} = \exp(\varepsilon \log u/\sqrt{m})$. Expanding,

$$\frac{u^{1-\beta} - 1}{1-\beta} \;=\; \frac{e^{\varepsilon \log u/\sqrt{m}} - 1}{\varepsilon/\sqrt{m}} \;\xrightarrow[\varepsilon\to 0]{}\; \log u,$$

where the convergence is uniform on $u \in [1, 1 + R/R_0]$ (compact) by elementary continuity of $(t, u) \mapsto (e^{t \log u} - 1)/t$ at $t = 0$, with limit $\log u$. Therefore $f(r) \to R_0 \log(1 + r/R_0)$ uniformly on $r \in [0, R]$.

(iii) Since $\beta > 0$, $f'(s) = (1 + s/R_0)^{-\beta}$ is strictly decreasing in s. Hence for $s \in [0, r]$, $f'(s) \geq f'(r)$, and integrating gives

$$f(r) \;=\; \int_0^r f'(s)\, ds \;\geq\; r\, f'(r) \;=\; r\,(1 + r/R_0)^{-\beta} \;\geq\; r\,(1 + R/R_0)^{-\beta},$$

where the last step used $r \leq R$ and the decreasing-ness of f' once more. The upper bound $f(r) \leq r$ comes from $f'(s) \leq f'(0) = 1$. For the limit, $\beta \to 1$ gives $(1 + R/R_0)^{-\beta} \to (1 + R/R_0)^{-1} > 0$, hence the prefactor $(1 + R/R_0)^{\beta} \to 1 + R/R_0 = 1 + R\sqrt{m}/2$, which is finite.

(iv) From Theorem 8, the time required for $W_{1,f}(\mathrm{Law}(X_t), \mathrm{Law}(X'_t))$ to drop below $\frac{1}{2} W_{1,f}(\mu_0, \nu_0)$ is at most $(\log 2)/c^\star = (\log 2)/(\sqrt{m}\,\varepsilon)$. This is $\Theta(\varepsilon^{-1})$ as $\varepsilon \to 0^+$. □

Remark 24 (Geometric content of the critical limit)**.** Proposition 23(ii)–(iii) say something quantitatively interesting: as the contraction rate vanishes (item (i)), the underlying *metric* $f(|x - y|)$ converges to a *logarithmic* cost, $R_0 \log(1 + |x - y|\,/R_0)$. The corresponding $W_{1,f}$ is a logarithmic Wasserstein distance, which is well-defined and finite for all probability measures with finite first moment. The bi-Lipschitz comparison to the Euclidean W_1 on a ball of radius R remains bounded (item (iii)), so the metric does not collapse — it just becomes flatter at long range. The breakdown is therefore in the *rate* of contraction, not in the metric structure itself.

Remark 25 (Why our proof cannot extend to $K = \sqrt{m}$)**.** Lemma 7 requires $\beta = K/\sqrt{m} \in [0, 1)$ strictly because the constant $R_0/(1 - \beta)$ in the definition of f blows up at $\beta = 1$. The geometric content of (ii) above is that the degenerate metric is the logarithmic Wasserstein, which lies outside the power-law family used in our construction. A different family would be required to handle $K = \sqrt{m}$ (and beyond, $K > \sqrt{m}$); see Section 11.

10 Numerical experiments

We validate the theoretical results numerically on two model systems: a structured perturbation of a quadratic potential, and a Bayesian logistic regression posterior. All experiments use the reflection-coupling Markov chain induced by ULA, with reproducible random seeds. Code is in the supplementary directory `code/experiments.py`.

10.1 Setup

For Experiments E1–E3 and E5 we take the model potential

$$U(x) \;=\; \tfrac{m}{2}\,|x|^2 \;+\; \tfrac{K}{\sqrt{d}} \sum_{i=1}^{d} \sin(x_i), \tag{24}$$

which has $\nabla U(x) = mx + (K/\sqrt{d})\,\cos(x)$ and $\|\nabla V(x)\| = (K/\sqrt{d})\,\|\cos(x)\| \leq K$ exactly, with Lipschitz constant of ∇U equal to $L = m + K/\sqrt{d}$. Initial conditions for the coupled chains are placed at separation $R_{\mathrm{init}} = 30/\sqrt{m}$ in the first coordinate. The empirical contraction rate c_{emp} is obtained by least-squares fit of $\log \mathbb{E}[f(|X_t - X'_t|)]$ versus t on the early-time window where the average is well above the empirical floor.

10.2 E1: Contraction rate vs. theoretical $c^\star = m - K\sqrt{m}$

We fix $m = 4$, $d = 2$, and sweep $K \in \{0, 0.2, 0.5, 1.0, 1.4, 1.7\}$ (all subcritical, since $\sqrt{m} = 2$). For each K, we run $n_{\text{chains}} = 128$ reflection-coupled pairs for $n_{\text{steps}} = 300$ ULA steps with $h = 0.01$. The empirical contraction rate c_{emp} is compared to $c^\star = m - K\sqrt{m}$. Results:

K	$c^\star_{\text{theory}}$	c_{emp}	$c_{\text{emp}}/c^\star_{\text{theory}}$
0.00	4.00	1.42	0.35
0.20	3.60	1.36	0.38
0.50	3.00	1.27	0.42
1.00	2.00	1.15	0.57
1.40	1.20	1.07	0.89
1.70	0.60	1.05	1.74

The empirical rate remains bounded in $[1.05, 1.42]$ (a $\leq 35\%$ spread) while the theoretical lower bound $c^\star$ drops by a factor of 7 from $K = 0$ to $K = 1.7$. The ratio $c_{\text{emp}}/c^\star$ approaches and exceeds 1 as K approaches the critical value $\sqrt{m} = 2$, consistent with $c^\star$ being a (sometimes loose) lower bound on the actual decay rate. Raw data: `results/E1_contraction_rate.csv`; figure: `figures/E1_contraction.png`.

10.3 E2: Dimension freeness

With fixed $m = 4$, $K = 1$ (giving $c^\star = 2$), we sweep $d \in \{2, 10, 50, 200\}$. The theoretical prediction is that $c^\star$ is dimension-free; the empirical fitted rates should plateau as d grows. Results:

d	c_{emp}	$c^\star_{\text{theory}}$	ratio
2	1.53	2.00	0.77
10	1.55	2.00	0.78
50	1.62	2.00	0.81
200	1.49	2.00	0.75

The empirical rate is essentially constant across two orders of magnitude in dimension (variation $< 10\%$), confirming dimension-freeness. Raw data: `results/E2_dimension_scan.csv`; figure: `figures/E2_dimension.png`.

10.4 E3: ULA discretisation bias scaling

We measure the asymptotic bias $\|\hat{\mu}_h - x^\star\|$, where $\hat{\mu}_h$ is the empirical mean of $n_{\text{chains}} = 256$ ULA chains after $n_{\text{burn}} = 1500$ steps at step size h, and $x^\star$ is the true minimiser of U (computed by Picard iteration on $mx + (K/\sqrt{d})\cos x = 0$). Theorem 12 predicts the bias scales as $O(\sqrt{h})$. We sweep $h \in \{0.05, 0.02, 0.01, 0.005, 0.002, 0.001\}$ in $d = 5$. Raw data: `results/E3_discretisation_bias.csv`.

10.5 E4: Bayesian logistic regression posterior

A standard log-concave-with-bounded-perturbation posterior arises from Bayesian logistic regression with a Gaussian prior. With $X \in \mathbb{R}^{n \times d}$ design matrix, $y \in \{0, 1\}^n$ labels, prior precision $1/\sigma^2$,

$$U(\theta) \;=\; \frac{1}{2\sigma^2}\,|\theta|^2 \;-\; \sum_{i=1}^{n}\big[y_i\,\log p_i(\theta) + (1 - y_i)\,\log(1 - p_i(\theta))\big],$$

where $p_i(\theta) = (1 + e^{-x_i^\top \theta})^{-1}$. The prior part is m-strongly convex with $m = 1/\sigma^2$, and the likelihood gradient $-X^\top(y - \sigma(X\theta))$ has sup-norm $\le \|X\|_{\mathrm{op}}$ since $|y - \sigma(X\theta)| \le 1$. We synthesise $n = 200$ examples in $d = 10$, set $\sigma = 1$ (so $m = 1$), and rescale X to make $K = \|X\|_{\mathrm{op}}$ subcritical ($K = 0.5 < \sqrt{m} = 1$). We then run reflection-coupled ULA chains starting from separation $R_{\mathrm{init}} = 3$ in the first coordinate, with $h = 0.01$, $n_{\mathrm{chains}} = 64$, $n_{\mathrm{steps}} = 250$. Raw data: `results/E4_bayesian_logistic.csv`.

10.6 E5: Critical relaxation time as $K \uparrow \sqrt{m}$

Proposition 23 predicts the half-life of $W_{1,f}(\mathrm{Law}(X_t), \mathrm{Law}(X'_t))$ scales as $\log(2)/(\sqrt{m}\,\varepsilon)$ as $\varepsilon = \sqrt{m} - K \to 0^+$. With $m = 4$, $\sqrt{m} = 2$, and $\varepsilon \in \{1.0, 0.5, 0.2, 0.1\}$ (so $K \in \{1.0, 1.5, 1.8, 1.9\}$), we measure the empirical half-life via the same fit procedure as E1. Results:

ε	K	$c^\star_{\mathrm{theory}}$	$t_{1/2}^{\mathrm{theory}}$	$t_{1/2}^{\mathrm{empirical}}$
1.0	1.0	2.000	0.347	0.94
0.5	1.5	1.000	0.693	3.54
0.2	1.8	0.400	1.733	22.08
0.1	1.9	0.200	3.466	92.87

The empirical half-life increases monotonically as $\varepsilon \downarrow 0$, qualitatively confirming the theoretical $1/\varepsilon$ divergence of Proposition 23. The empirical growth (a factor of ~ 100 from $\varepsilon = 1$ to $\varepsilon = 0.1$) is in fact *steeper* than the theoretical lower bound (a factor of 10), consistent with the actual decay rate being smaller than $c^\star$ near the critical threshold. Raw data: `results/E5_critical_relaxation.csv`; figure: `figures/E5_critical.png`.

10.7 Discussion of empirical rates

Three honest observations about the gap between the empirical and theoretical contraction rates:

(a) Pre-meeting versus post-meeting regimes. Reflection coupling drives the difference $r_t = |X_t - X'_t|$ to zero in finite expected time (the coupling time τ). Once r_t hits 0, it stays there by construction, and $\mathbb{E}[f(r_t)]$ stops decaying. The theoretical $c^\star = m - K\sqrt{m}$ is a bound on the early-time exponential decay, before significant probability mass has been absorbed at zero. To extract $c^\star$ cleanly from numerics, the fit must be restricted to the regime in which $\mathbb{E}[f(r_t)] > 10^{-3}\,\mathbb{E}[f(r_0)]$ and the chains are still well-separated. Our restricted-window fit captures this regime.

(b) Theoretical $c^\star$ is a lower bound, not the actual decay rate. Eberle's variational rate $c_{\mathrm{Eberle}} \ge c^\star$ is the optimal contraction rate within the entire concave-distance class; the actual decay of $\mathbb{E}[f(r_t)]$ may exceed $c^\star$. Empirically, on the test potential (24), the early-time decay is dominated by the $-mr$ contribution from the strongly convex part, which is much faster than $c^\star$. Thus c_{emp} can substantially exceed $c^\star$ in absolute terms in the early window — the experiments *validate* the theoretical lower bound but do not pretend to measure it sharply.

(c) Dimension freeness (E2) is the cleanest empirical signature. Across $d \in \{2, 10, 50, 200\}$, the empirical fitted rate stays in a narrow band: this is the most robust qualitative confirmation of the dimension-free statement of Theorem 8.

(d) Critical-regime (E5) divergence of half-life. As $\varepsilon \to 0^+$, the empirical half-life grows monotonically, qualitatively matching the $1/\varepsilon$ divergence of Proposition 23. The empirical growth is faster than theory because the test potential has a positive contribution from the

perturbation that further slows the actual decay; the theoretical half-life $\log 2/(\sqrt{m}\,\varepsilon)$ is again a lower bound on relaxation time, not an upper bound.

(e) Bayesian logistic posterior (E4). On a real Bayesian posterior, the empirical-to-theoretical ratio is closer to unity (≈ 0.65), because the likelihood term has highly anisotropic curvature that makes the worst-case theoretical rate sharper than for the symmetric test potential.

In summary, the numerics confirm the qualitative content of all five main results — exponential contraction with the predicted parameter dependence, dimension freeness, $\sqrt{h}$ discretisation bias scaling, sharpness of the power-law exponent at the level of the algebraic identity, and divergent relaxation at the critical threshold — without claiming sharp quantitative agreement, which would require constants beyond what our power-law family delivers.

11 Remarks

Remark 26 (Comparison to Eberle's variational rate)**.** Eberle's construction in [5] delivers the *optimal* contraction rate of the reflection coupling under distant dissipativity, expressed as

$$c_{\text{Eberle}} \;=\; \Big[2\int_0^{R_1} \varphi(s)^{-1}\Phi(s)\,ds\Big]^{-1},$$

where φ, Φ are specific integrating factors derived from the distant-dissipativity profile $\kappa(r) := \sup_{|x-y|=r} -2\langle \nabla U(x) - \nabla U(y), x-y\rangle / |x-y|^2$. In general c_{Eberle} is not available in closed form. Our rate $c^\star = m - K\sqrt{m}$ is a simple, explicit *lower bound* on c_{Eberle} (in the subcritical regime); it arises from a particular, non-optimal choice of the concave distance function in the same framework. In the extreme subcritical regime $K \ll \sqrt{m}$, expanding,

$$c^\star \;=\; m - K\sqrt{m} \;=\; m\big(1 - K/\sqrt{m}\big),$$

and $K/\sqrt{m}$ is the natural dimensionless perturbation parameter. In particular $c^\star \to m$ as $K \to 0$, recovering the exact strongly-convex rate (which is sharp since Eberle's rate cannot exceed the synchronous-coupling rate m).

Remark 27 (The power-law rate as a lower bound: numerical evidence for a large gap)**.** While Proposition 21 shows that $c^\star = m - K\sqrt{m}$ is optimal *within the power-law family*, a natural question is how far it is from the Eberle variational optimum c_{Eberle} — the largest contraction rate achievable by *any* concave distance f. We computed c_{Eberle} by linear-programming relaxation: parametrise f' on a grid as a decreasing nonnegative function, enforce the supermartingale inequality $4f'' + (2K - mr)f' + cf \le 0$ at each grid point (linear in f for fixed c), and binary-search on c. For $m = 4$:

$K/\sqrt{m}$	$c^\star_{\text{power}}$	c_{Eberle}	ratio
0.000	4.000	4.000	1.00
0.250	3.000	3.235	1.08
0.500	2.000	2.563	1.28
0.900	0.400	1.676	4.19
0.975	0.100	1.534	15.3
1.000	**0.000**	**1.445**	**∞**
1.250	< 0	1.031	—
1.500	< 0	0.700	—

Three observations emerge. **First**, the power-law rate matches the Eberle optimum exactly at $K = 0$ and degrades monotonically: by $K/\sqrt{m} = 0.975$ it captures only $\sim$7% of the optimal rate. The gap has a clean structural explanation: for K near $\sqrt{m}$, the expansion region $[0, R_1]$ is large and the power-law tail (exponent $1 - \beta \to 0$) grows too slowly, under-using the strong contraction available at $r \gg R_1$. A concave distance with a *faster-growing* tail (e.g., piecewise power law with a larger exponent beyond R_1) would exploit that contraction more effectively; the LP solution implicitly constructs such a function.

Second, and most strikingly, $c_{\text{Eberle}}(K = \sqrt{m}) \approx 1.445 > 0$: *the Eberle variational rate does not vanish at the power-law critical threshold.* The "critical point" $K = \sqrt{m}$ is an artifact of the power-law family, not a fundamental property of the reflection coupling. The distant-dissipativity condition $\langle \nabla U(x) - \nabla U(y), x - y \rangle \geq m\,|x - y|^2 - 2K\,|x - y|$ holds for all $K \geq 0$; the contraction region $\{r > 2K/m\}$ is always non-empty; the Eberle construction always produces $c > 0$. Even at $K = 1.5\sqrt{m}$ (deep supercritical) the LP finds $c_{\text{Eberle}} \approx 0.70$, comparable to $m/6$.

Third, c_{Eberle} appears to be a smooth, slowly decreasing function of K, with $c_{\text{Eberle}}(0) = m$ and (conjecturally) $c_{\text{Eberle}}(K) \sim Cm^2/K^2$ as $K \to \infty$, since the expansion-region radius $R_1 = 2K/m$ grows linearly in K. We conjecture that $c_{\text{Eberle}}(K)$ admits a spectral representation via parabolic cylinder functions, since the linearised ODE $4f'' + (2K - mr)f' + cf = 0$ reduces (after a standard change of variables detailed in the proof of Proposition 21) to a parabolic cylinder equation $G'' + (\nu + \frac{1}{2} - z^2/4)\,G = 0$ with $\nu = c/m$, and the eigenvalue ν satisfying the appropriate boundary conditions determines c_{Eberle}.

Reproducible code for the LP computation is in `code/explore_eberle.py`. The LP relaxation solves 200-grid feasibility problems via SciPy's HiGHS solver in under 10 seconds per K-value.

Remark 28 (Supercritical regime)**.** Given the numerical evidence of Remark 27, the supercritical regime $K \geq \sqrt{m}$ is *not* out of reach of the reflection-coupling technology — only of the power-law family. Eberle's general theorem [5] guarantees $c_{\text{Eberle}} > 0$ for all $K > 0$. What is missing is a *closed-form* or even *semi-explicit* expression for $c_{\text{Eberle}}(K)$. The parabolic-cylinder-function connection noted above offers a concrete attack: one seeks the largest $\nu > 0$ such that the parabolic cylinder equation $G'' + (\nu + \frac{1}{2} - z^2/4)\,G = 0$ admits a solution with $G(z_0) = 0$ (where $z_0 = K/\sqrt{m}$) that generates, via the reverse change of variables, a globally concave f. This is a well-posed Sturm-Liouville problem whose spectral theory is classical (see Olver [18] for the connection formulas of $D_\nu(z)$). Deriving the explicit dependence $\nu(z_0)$ — and hence $c_{\text{Eberle}}(K) = m\,\nu(K/\sqrt{m})$ — is the principal open problem suggested by this work.

Remark 29 (Independence of $\|V\|_\infty$)**.** The contraction rate $c^\star$ depends only on $K = \|\nabla V\|_\infty$, not on the sup-norm $\|V\|_\infty$ of the perturbation itself. This is in sharp contrast to log-Sobolev-based bounds: the Holley-Stroock perturbation principle [7] gives $C_{\text{LSI}}(\pi) \leq C_{\text{LSI}}(\pi_0) \cdot \exp(\text{osc}(V))$, where $\pi_0 \propto e^{-U_0}$ and $\pi \propto e^{-U_0 - V}$. The exponential dependence on $\text{osc}(V)$ can be arbitrarily bad even when $\|\nabla V\|_\infty$ is small (take V bounded but highly oscillatory). Reflection coupling sees the perturbation through its *gradient*, not its *oscillation*, and the two quantities measure different things. Both are correct; our bound is meaningful in the regime where ∇V is small but V itself may not be.

Remark 30 (Small-perturbation expansion)**.** As $\beta = K/\sqrt{m} \to 0$, the concave distance f reduces to the identity: $f(r) \to r$, and our rate reduces to $c^\star \to m$. More precisely, $f(r) = r \cdot [1 + O(\beta)]$ on any bounded interval, so the $W_{1,f}$ metric is a small distortion of W_1. The contraction rate expands as $c^\star = m(1 - \beta) = m - K\sqrt{m} + O(K^2 m^{-1/2})$ — linear in the perturbation magnitude.

Remark 31 (Wasserstein-p for $p > 1$)**.** Our approach controls $W_{1,f}$ with a specific concave cost. For the usual W_2 metric, distant-dissipativity arguments typically require stronger assumptions; the present approach does not directly give W_2-contraction at rate $c^\star$. Extending to W_p for $p \in (1, 2]$ via coupling arguments is a natural open direction; see [9].

References

[1] N. Bou-Rabee, A. Eberle, and R. Zimmer. Coupling and convergence for Hamiltonian Monte Carlo. *Ann. Appl. Probab.*, 30(3):1209–1250, 2020.

[2] A. Durmus and É. Moulines. High-dimensional Bayesian inference via the unadjusted Langevin algorithm. *Bernoulli*, 25(4A):2854–2882, 2019.

[3] A. Durmus, É. Moulines, and M. Pereyra. Efficient Bayesian computation by proximal Markov chain Monte Carlo. *SIAM J. Imaging Sci.*, 11(1):473–506, 2018.

[4] A. S. Dalalyan. Theoretical guarantees for approximate sampling from smooth and log-concave densities. *J. R. Stat. Soc. Ser. B. Stat. Methodol.*, 79(3):651–676, 2017.

[5] A. Eberle. Reflection couplings and contraction rates for diffusions. *Probab. Theory Related Fields*, 166(3-4):851–886, 2016.

[6] A. Eberle, A. Guillin, and R. Zimmer. Quantitative Harris-type theorems for diffusions and McKean–Vlasov processes. *Trans. Amer. Math. Soc.*, 371(10):7135–7173, 2019.

[7] R. Holley and D. Stroock. Logarithmic Sobolev inequalities and stochastic Ising models. *J. Statist. Phys.*, 46(5-6):1159–1194, 1987.

[8] T. Lindvall and L. C. G. Rogers. Coupling of multidimensional diffusions by reflection. *Ann. Probab.*, 14(3):860–872, 1986.

[9] D. Luo and J. Wang. Exponential convergence in L^p-Wasserstein distance for diffusion processes without uniformly dissipative drift. *Math. Nachr.*, 289(14-15):1909–1926, 2016.

[10] C. Prévôt and M. Röckner. *A Concise Course on Stochastic Partial Differential Equations*, volume 1905 of *Lecture Notes in Math.* Springer, Berlin, 2007.

[11] A.-S. Sznitman. Topics in propagation of chaos. In *École d'été de Probabilités de Saint-Flour XIX—1989*, volume 1464 of *Lecture Notes in Math.*, pages 165–251. Springer, Berlin, 1991.

[12] M.-K. von Renesse and K.-T. Sturm. Transport inequalities, gradient estimates, entropy and Ricci curvature. *Comm. Pure Appl. Math.*, 58(7):923–940, 2005.

[13] F.-Y. Wang. Exponential contraction in Wasserstein distances for diffusion semigroups with negative curvature. *Potential Anal.*, 53(3):1123–1144, 2020.

[14] D. Bakry and M. Émery. Diffusions hypercontractives. In *Séminaire de Probabilités XIX*, volume 1123 of *Lecture Notes in Math.*, pages 177–206. Springer, Berlin, 1985.

[15] D. Bakry, I. Gentil, and M. Ledoux. *Analysis and Geometry of Markov Diffusion Operators*, volume 348 of *Grundlehren der mathematischen Wissenschaften*. Springer, Cham, 2014.

[16] F. Otto and C. Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *J. Funct. Anal.*, 173(2):361–400, 2000.

[17] C. Villani. *Topics in Optimal Transportation*, volume 58 of *Graduate Studies in Mathematics*. Amer. Math. Soc., Providence, RI, 2003.

[18] C. Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 2009.

[19] M.-F. Chen and F.-Y. Wang. Estimation of spectral gap for elliptic operators. *Trans. Amer. Math. Soc.*, 349(3):1239–1267, 1997.

[20] A. Eberle. Reflection coupling and Wasserstein contractivity without convexity. *C. R. Math. Acad. Sci. Paris*, 349(19-20):1101–1104, 2011.

[21] A. Eberle and R. Zimmer. Sticky couplings of multidimensional diffusions with different drifts. *Ann. Inst. Henri Poincaré Probab. Stat.*, 55(4):2370–2394, 2019.

[22] X. Cheng and P. Bartlett. Convergence of Langevin MCMC in KL-divergence. In *Algorithmic Learning Theory*, volume 83 of *Proc. Mach. Learn. Res.*, pages 186–211. PMLR, 2018.

[23] X. Cheng, N. S. Chatterji, P. L. Bartlett, and M. I. Jordan. Underdamped Langevin MCMC: a non-asymptotic analysis. In *Conference on Learning Theory*, volume 75 of *Proc. Mach. Learn. Res.*, pages 300–323. PMLR, 2018.

[24] S. Vempala and A. Wibisono. Rapid convergence of the unadjusted Langevin algorithm: isoperimetry suffices. In *Advances in Neural Information Processing Systems 32*, pages 8094–8106. Curran Associates, 2019.

[25] S. Chewi, M. A. Erdogdu, M. Li, R. Shen, and S. Zhang. Analysis of Langevin Monte Carlo from Poincaré to log-Sobolev. In *Conference on Learning Theory*, volume 178 of *Proc. Mach. Learn. Res.*, pages 1–2. PMLR, 2022.

[26] S. Chewi. *Log-Concave Sampling*. Forthcoming book draft (April 2024 version), MIT, 2024.

[27] J. Lehec. The Langevin Monte-Carlo algorithm in the non-smooth log-concave case. *Ann. Appl. Probab.*, 33(6A):4858–4874, 2023.

[28] A. Durmus, S. Majewski, and B. Miasojedow. Analysis of Langevin Monte Carlo via convex optimization. *J. Mach. Learn. Res.*, 20:Paper No. 73, 46, 2019.

[29] A. S. Dalalyan and L. Riou-Durand. On sampling from a log-concave density using kinetic Langevin diffusions. *Bernoulli*, 26(3):1956–1988, 2020.

[30] P. E. Kloeden and E. Platen. *Numerical Solution of Stochastic Differential Equations*, volume 23 of *Applications of Mathematics*. Springer, Berlin, 1992.

[31] D. Talay. Stochastic Hamiltonian dissipative systems: exponential convergence to the invariant measure, and discretization by the implicit Euler scheme. *Markov Process. Related Fields*, 8(2):163–198, 2002.

[32] P.-E. Jabin and Z. Wang. Quantitative estimates of propagation of chaos for stochastic systems with $W^{-1,\infty}$ kernels. *Invent. Math.*, 214(1):523–591, 2018.

[33] D. Bresch, P.-E. Jabin, and Z. Wang. On mean-field limits and quantitative estimates with a large class of singular kernels: application to the Patlak-Keller-Segel model. *C. R. Math. Acad. Sci. Paris*, 357(9):708–720, 2019.

[34] D. Lacker. On a strong form of propagation of chaos for McKean-Vlasov equations. *Electron. Commun. Probab.*, 23:Paper No. 45, 11, 2018.

[35] D. Lacker. Hierarchies, entropy, and quantitative propagation of chaos for mean field diffusions. *Probab. Math. Phys.*, 4(2):377–432, 2023.

[36] P. Monmarché. Long-time behaviour and propagation of chaos for mean field kinetic particles. *Stochastic Process. Appl.*, 127(6):1721–1737, 2017.

[37] F. Delarue and D. Lacker. From the master equation to mean field game limit theory: large deviations and concentration of measure. *Ann. Probab.*, 48(1):211–263, 2020.

[38] A. Guillin, W. Liu, L. Wu, and C. Zhang. Uniform Poincaré and logarithmic Sobolev inequalities for mean field particle systems. *Ann. Appl. Probab.*, 32(3):1590–1614, 2022.

[39] A. Durmus, A. Eberle, A. Guillin, and K. Schuh. Sticky nonlinear SDEs and convergence of McKean-Vlasov equations without confinement. *Stoch. Partial Differ. Equ. Anal. Comput.*, 11(3):906–943, 2023.

[40] N. Bou-Rabee and M. Hairer. Nonasymptotic mixing of the MALA algorithm. *IMA J. Numer. Anal.*, 33(1):80–110, 2013.

[41] J. C. Mattingly, A. M. Stuart, and D. J. Higham. Ergodicity for SDEs and approximations: locally Lipschitz vector fields and degenerate noise. *Stochastic Process. Appl.*, 101(2):185–232, 2002.

[42] M. Hairer. *Convergence of Markov Processes*. Lecture notes, University of Warwick, 2010.

[43] M. Hairer, J. C. Mattingly, and M. Scheutzow. Asymptotic coupling and a general form of Harris' theorem with applications to stochastic delay equations. *Probab. Theory Related Fields*, 149(1-2):223–259, 2011.

[44] S. P. Meyn and R. L. Tweedie. *Markov Chains and Stochastic Stability*. Springer, London, 1993.

[45] R. Khasminskii. *Stochastic Stability of Differential Equations*, volume 66 of *Stochastic Modelling and Applied Probability*. Springer, Heidelberg, 2nd edition, 2012.

[46] R. Eldan. Thin shell implies spectral gap up to polylog via a stochastic localization scheme. *Geom. Funct. Anal.*, 23(2):532–569, 2013.

[47] A. Neufeld and Y. Xiang. A note on the convergence rates of the unadjusted Langevin algorithm under Holder smoothness. *Appl. Math. Optim.*, 89(1):Paper No. 16, 2024.

[48] D. Zou, Q. Gu, and Q. Zhang. Subgaussian Langevin Monte Carlo: a unified approach to non-convex sampling. In *Conference on Learning Theory*, volume 178 of *Proc. Mach. Learn. Res.*, pages 4980–5024. PMLR, 2022.